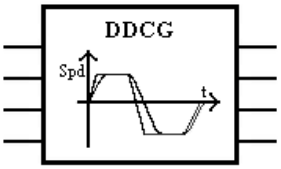
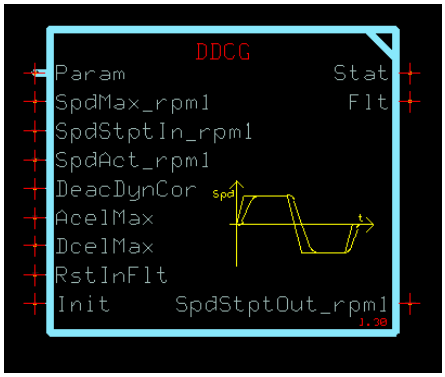
DDCG





Function:

To obtain a comfortable drive feeling the speed curve is generated with a drive curve generator dependent on the variation of the set value. Beside the actual speed, it is also considered how far and how fast the set value is changed. According to the variation of the set value a very sensitive and smooth drive feeling can also be performed as well as a dynamic driving.

Inputs/Outputs

Inputs

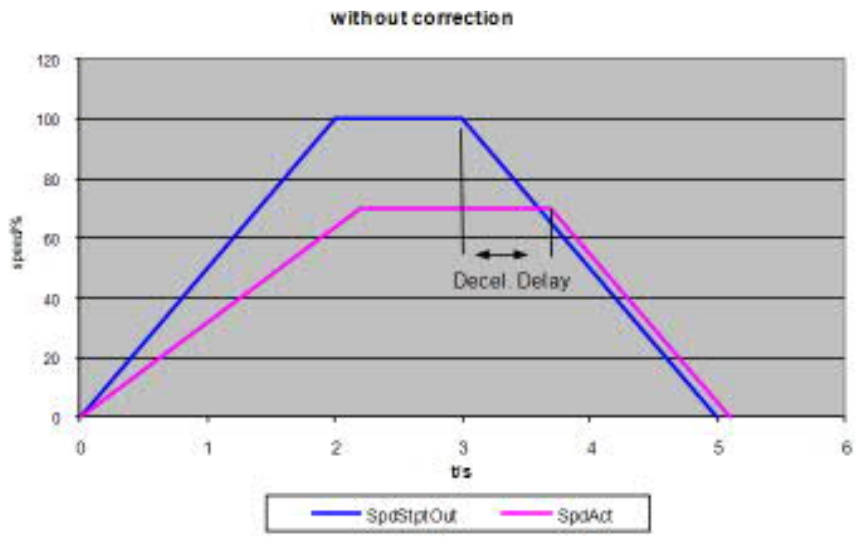
Item/Range	Description
SpdStptIn_rpm1 (S32): (-SpdMax_rpm1….SpdMax_rpm1 [0.1 rpm])	Speed set value.
SpdAct_rpm1 (S32) (-200000….200000 [0.1 rpm])	Actual speed value (motor)
SpdMax_rpm1(U32): (1…200000 [0.1 rpm])	Max. speed value of the application.
AcelMax (S32): (1….500000 [0.1 rpm/s])	**Acceleration:** Specifies the max. acceleration, without rounding and without progression (Roundn.=0, Progr.=0). The maximum acceleration is achieved, if the set value escalates from zero to maximum speed.
DecelMax (S32) (1….500000 [0.1 rpm/s])	**Deceleration:** Specifies the max. deceleration, without rounding and without progression (R=0, P=0). The maximum deceleration is achieved, if the set value jumps down from maximum speed to zero.
Init (BOOL)	Init. function of the dynamic correction. (see DeacDynCor) (0….1; pos. transition)
DeacDynCor (BOOL)	Deactivation of dynamic correction of drive curve output value: At inversion from acceleration to deceleration or at a pos. transition at "Init" input, a correction of the drive curve is made, the drive curve output value is immediately set to the actual speed value (SpdAct_rpm1), see Graph DDCG_1 and Graph DDCG_2. (0 = dynamic correction; 1 = no dyn.- correction)

Inputs/Outputs (continued)

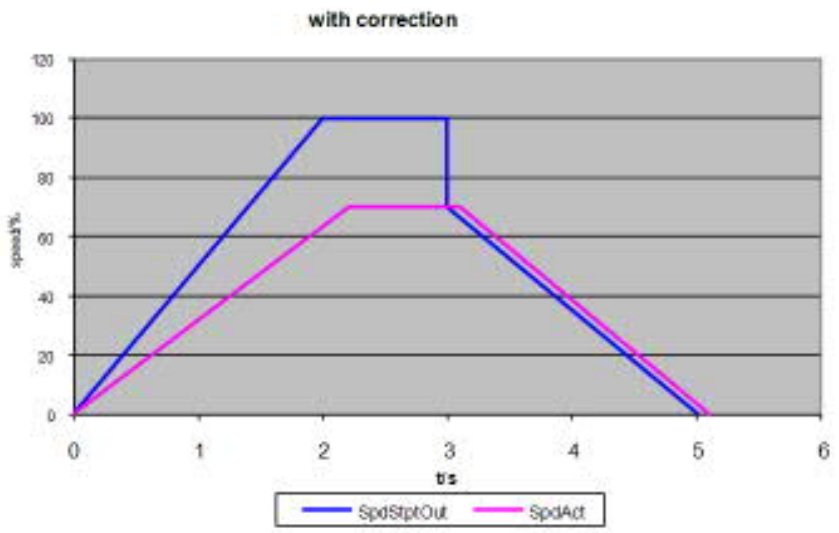
Inputs

Item/Range	Description
RstInFlt (BOOL)	Reset of the input faults TRUE: Fault is not latched FALSE: Fault is latched until fixed and power is cycled

Graph DDCG_1



Graph DDCG_2



Inputs/Outputs (continued)

Output

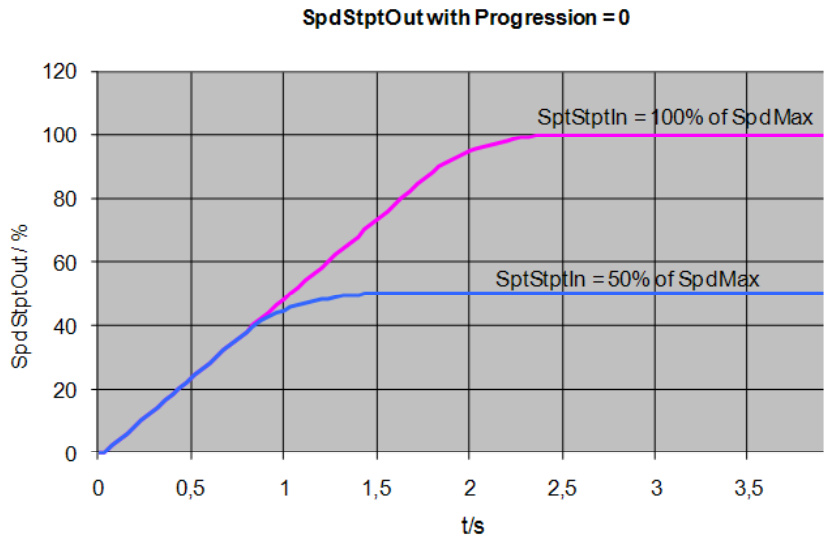
Item/Range	Description
SpdStptOut_rpm1 (S32): (-20000….200000) [0.1 rpm]	Calculated speed set value.
Stat (U16)	Status code of the function block
Flt (U16)	Fault code of the function block

Parameters

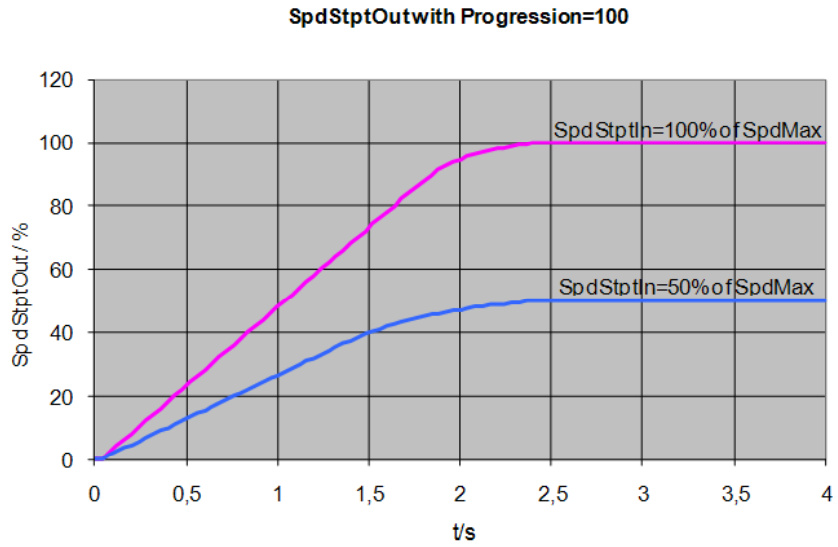
Item/Range	Description
Progression (U8) (0…100%; default value: 0)	Specifies the influence of the set value variation to the value of acceleration or deceleration. **Progression = 0:** Independently from set value variation always the maximum acceleration or deceleration is active. (Graph DDCG_3) **Progression = 100:** The value of acceleration or deceleration depends on the variation of the set value on a linear characteristic. (Graph DDCG_4)
Roundness (U8) (0…90; default value: 0)	Specifies the strength of the roundness (0: no rounding; 90: max. rounding). The speed curve is always rounded before reaching the final value. A rounding at the beginning of acceleration or deceleration becomes effective only if the set value changes slowly. With a slow change of the set value a smooth drive reaction is performed, as acceleration is increased continuously. In case of a fast change of the set value an aggressive drive reaction results, as acceleration achieves its maximum value precipitously. (Graph DDCG_5)
FltDetTm_ms (U16) (0…1000 [ms]; default value: 0)	Fault detection time The time a fault needs to be present before it is declared

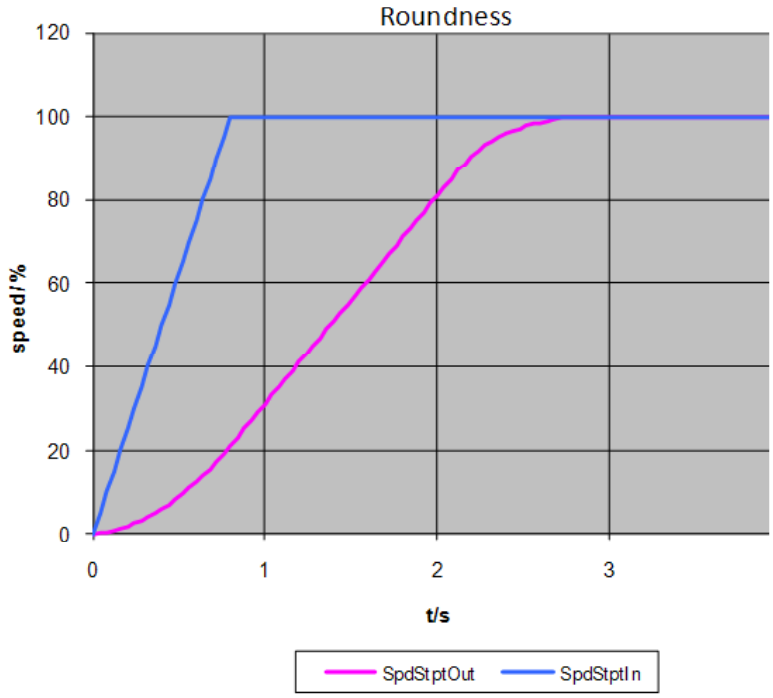
Inputs/Outputs (continued)

Graph DDCG_3



Graph DDCG_4



Inputs/Outputs (continued)

Graph DDCG_5